



# Techno-economic analysis and viability assessment of concentrating solar power under climatic conditions of Rwanda

M. C. Mukeshimana<sup>1</sup> · Z.-Y. Zhao<sup>1</sup> · J. P. Nshimiyimana<sup>2</sup>

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## Abstract

This study performs a techno-economic analysis of concentrated solar power (CSP) in Rwanda, by modelling two technologies, solar tower power plant (STPP) and parabolic trough power plant (PTPP). A 100 M plant for each technology was simulated at two different locations (Nyanza and Kayonza) using system advisor model (SAM) software. The main objective is to obtain suitable designs for the selected sites and to evaluate the viability of CSP in Rwanda, as no such study has been done so far. Optimization of solar field size and thermal energy storage (TES) is performed to obtain minimum levelized cost of electricity (LCOE). Results showed that solar multiple (SM) of 2.8 with TES of 8 h were the optimal configuration for the STPP, and SM of 3.2 with TES of 6 h for the PTPP. The LCOE under the optimized configurations could then be minimized to 11.33 ¢/kWh and 12.92 ¢/kWh for the STPP at Nyanza and Kayonza, respectively; and 10.51 ¢/kWh and 11.30 ¢/kWh for the PTPP. Moreover, sensitivity analysis shows that the financing conditions significantly affect the economic feasibility of CSP in Rwanda. Finally, major barriers hindering the development of CSP are highlighted, together with policy recommendations. This study may assist policymakers to make appropriate decisions during CSP development in Rwanda and many other countries in Africa.

**Keywords** Concentrating solar power · System advisor model · Solar tower · Parabolic trough · Rwanda

## Abbreviations

|             |                                          |
|-------------|------------------------------------------|
| AEG         | Annual net electricity generation (kWh)  |
| $A_{sf}$    | Solar field area (m <sup>2</sup> )       |
| CF          | Capacity factor (%)                      |
| CSP         | Concentrating solar power                |
| $d$         | Discount rate (%)                        |
| DNI         | Direct Normal Irradiance                 |
| $E_{gross}$ | Gross power output (kWh)                 |
| $E_{par}$   | Power output from parasitic losses (kWh) |
| $F_t$       | Fuel expenditures in year $t$ (\$)       |
| HTF         | Heat transfer fluid                      |
| IRR         | Internal rate of return (%)              |
| $I_t$       | Investment expenditures in year $t$ (\$) |

|               |                                                                              |
|---------------|------------------------------------------------------------------------------|
| LCOE          | Levelized cost of electricity (\$/kWh)                                       |
| $n$           | Project lifetime (years)                                                     |
| NPV           | Net present value (\$)                                                       |
| $OM_t$        | Operation and maintenance expenditures in year $t$ (\$)                      |
| PTPP          | Parabolic trough power plant                                                 |
| $Q_{ST}$      | Thermal energy output per unit square meter of collector (kWh <sub>i</sub> ) |
| RE            | Renewable energy                                                             |
| SAM           | System advisor model                                                         |
| SM            | Solar multiple                                                               |
| SSA           | Sub-Saharan Africa                                                           |
| STPP          | Solar tower power plant                                                      |
| TES           | Thermal energy storage                                                       |
| TMY           | Typical meteorological year                                                  |
| $\eta_{PB}$   | Power block efficiency                                                       |
| $\eta_{sh}$   | Shading efficiency (%)                                                       |
| $\eta_{soil}$ | Soiling effect (%)                                                           |
| $\eta_{sto}$  | Thermal energy storage efficiency (%)                                        |
| $\eta_{ta}$   | Turbine-alternator set efficiency                                            |

✉ M. C. Mukeshimana  
 muclairese@hotmail.com

✉ Z.-Y. Zhao  
 zhaozhenyuxm@263.net

<sup>1</sup> Beijing Key Laboratory of New Energy and Low-Carbon Development, School of Economics and Management, North China Electric Power University, Beijing 102206, China

<sup>2</sup> School of Science, College of Science and Technology, University of Rwanda, P.O. Box: 3900, Kigali, Rwanda

## Introduction

Rwanda, just like many countries in Sub-Saharan Africa (SSA), has been struggling to generate enough electricity to promote growth and development in both urban and rural areas, where only about a half of the population have access to electricity (IEA, 2020). The growing economy combined with its rapidly increasing and gradually more urbanized population have resulted in the imbalance between energy demand and supply. The total installed capacity is about 221 MW, from which thermal energy (diesel and heavy fuel generators) accounts 27% (Eustache et al., 2019; Mukeshimana et al., 2021b). Thermal energy is the most expensive source of power and is harmful to the environment.

The government of Rwanda is conscious that ensuring access to modern, sustainable, and affordable energy services is vital to the country's vision of economic development, poverty eradication and socioeconomic transformation (Republic of Rwanda, 2017). The country is ambitiously targeting a generation capacity of 556 MW and a 100% electrification rate by 2024, whereby 52% will have access to grid connection and 48% off-grid (MININFRA, 2018). As such, efforts are needed to tap more into renewable energy (RE) resources to achieve this target while at the same time reducing the consumption of fossil fuels. Currently, the country derives almost all of its RE from some hydropower plants, which occupy nearly a half of its electricity generation (Eustache et al., 2019; Uwisengeyimana et al., 2016). However, other RE technologies are at their early stage of development, despite their abundant potential (Munyaneza et al., 2016). Solar power technologies are mostly untapped, mainly due to the lack of information related to solar power resources and the negative perception that solar technologies are more costly compared to fossil fuel technologies. Currently, only photovoltaic (PV) technologies have been adopted and account about 5% (12 MW) of the country's total power generation (Eustache et al., 2019; MININFRA, 2018), whereas concentrating solar power (CSP) technologies have long been totally overlooked. It is vital to undertake feasibility analysis of any project prior to its implementation as it can provide enough information during the decision-making process about whether to proceed with the project or not.

For the past few years, solar power technologies have been attracting great attention worldwide (Nižetić & Peng, 2017; Penga et al., 2019). Currently, the global capacity of CSP plants is estimated at 9267 MW in 2020, among which 6128 MW are operational, 1547 MW are under construction, while 1592 MW are currently under development (SolarPaces, 2020). According to International Energy Agency (IEA), Africa is projected to be the

second-largest producer and by far the largest exporter of electricity from CSP by 2050 (IEA, 2010). This is a great opportunity for CSP project developers to start investing in CSP projects in African countries where CSP is suitable.

The concept of CSP involves the use mirrors or reflectors to concentrate or reflect the energy from the sun and collect it to heat a working fluid (heat transfer fluid, HTF) for thermal energy and/or electricity production. There are currently four commercial technologies available for CSP, namely parabolic trough (PT), solar tower (ST), linear Fresnel reflector and solar dish (Py et al., 2013; Shahzad Nazir et al., 2021; Zhang et al., 2013). Among these CSP technologies, PT and ST systems continue to dominate the market, with PT systems being the most mature of all, accounting more than 90% of the installed capacity (Baharoon et al., 2015; IEA, 2010). PT systems use parabolic-shaped troughs with reflective surfaces to focus the sun's energy onto a receiver tube that is installed at the focus of a trough. ST systems use a large field of flat mirrors, called heliostats, to direct the sun's energy onto a focal point of a single receiver mounted high on a central tower, which allows for higher operating temperatures and thus greater efficiencies. One of the main advantages of CSP is the fact that they can be integrated with thermal energy storage (TES) to operate even during cloudy skies or after sunset. This storage capacity of CSP contributes to the significant increase of the capacity factor (CF) and dispatchability of the generated electricity, thus improving the economic competitiveness of CSP plants (Gallo et al., 2016; IRENA, 2012; Kuravi et al., 2013; Lovegrove & Stein, 2012). To provide the required heat capacity, the solar field (i.e. concentrating mirrors and heat collectors) of the CSP plant must be oversized with respect to the gross output of the plant. However, oversizing the solar field leads to an increased capital cost of the plant as a result of increased installation costs of the components. To balance the tradeoff between the incremental cost associated with the increased solar field and the TES, it is always necessary to find optimum combination between the solar field size and TES capacity which gives a maximum utilization of the power block and at the same time minimizes the overall cost of the electricity generated by the CSP installations. The size of a solar field is defined in terms of an important parameter, the solar multiple (SM), which corresponds to the ratio of the actual size of a solar field to the size of the field required to operate the power cycle at its rated capacity. An  $SM = 1.0$  indicates that the solar field size generates the same amount of thermal energy required to drive the power block at its design gross output, while an  $SM > 1.0$  indicates that the solar field size can generate more thermal energy than the energy required to drive the power block at its design gross output. To cope with

thermal losses, plants with no storage typically have an SM between 1.1 and 1.5 while plants with thermal storage may have an SM between 2.0 and 5.0 (Lovegrove & Stein, 2012).

Several studies have been carried out to assess the techno-economic potential of CSP in many areas around the world (Agyekum & Velkin, 2020; Aly et al., 2019; Hirbodi et al., 2020; Ogunleye et al., 2020; Sultan et al., 2020; Suresh et al., 2014; Trabelsi et al., 2016). However, while extensive research has been done in many parts of the world, to the best of the authors' knowledge, only one study has recently attempted to investigate the feasibility of CSP in Rwanda (Hagumimana et al., 2021). Still, the study only focused on one type of CSP technology (solar tower) and has not taken into consideration the effect of various financial conditions on the overall performance of the CSP plant. To fill the existent literature gap, this study conducts a comprehensive techno-economic investigation to ascertain the viability of CSP technology and the feasibility of its implementation in Rwanda. The main contributions of the study are as follows: (1) determine suitable designs for the two different types of CSP technologies (parabolic trough and solar tower) in the most favorable sites of Rwanda; (2) evaluate the technical and economic performance of the proposed designs; (3) investigate the feasibility of CSP implementation in Rwanda through a sensitivity analysis. A comprehensive study on the techno-economic feasibility of CSP bridges the research gap on large-scale solar power in Rwanda and will particularly add value to the country's power planning sector. This study contributes to the new literature not only by better understanding the viability of CSP technology in Rwanda but also can serve as a guide in many other developing countries to incorporate CSP in their power generation mix.

## Materials and methods

### CSP modeling with system advisor model

The main reason of modeling a solar energy system is to evaluate the amount of energy it will supply over a long time. This would help in designing and optimizing its performance and cost and hence in determining the economic viability of the designed system. The modeling of parabolic trough power plant (PTPP) and solar tower power plant (STPP) was performed in this study using the System Advisor Model (SAM) software, version 2020.2.29. Developed by the National Renewable Energy Laboratory (NREL), in conjunction with Sandia National Lab and the US Department of Energy, SAM is a very powerful tool designed to facilitate decision making for RE system performance and economic analysis (Blair et al., 2018; Wagner & Gilman, 2011). SAM is commonly used to calculate performance and

financial metrics of various solar power plants at specified sites based on project costs and system design parameters (Dobos et al., 2014; Ezeanya et al., 2018). The SAM-based simulation modeling has been used and validated in a variety of RE systems, many of which were highlighted by (Ezeanya et al., 2018). This shows the reliability of SAM in designing and evaluating the techno-economic potential of CSP projects for a specified location. The projects modelled in this study include the CSP physical parabolic trough and CSP power tower molten salt as performance models, together with the Single Owner Power Purchase Agreement as a financial model.

The diagram displayed in Fig. 1 shows different steps of the simulation in SAM software. The model consists of different components, which in turn have several parameters that are used as input data to produce needed output parameters. These components are classified into three categories, namely the location and resources, technical design components and economic parameters. The location and resources components include geographical and meteorological data of the locations. The technical design components include the solar field, collectors, receivers, power cycle, and thermal storage for the PTPP; and system design, tower and receiver, heliostat field, power cycle and thermal storage for the STPP. They are mainly used to model the technical performance of the plants. As for the economic parameters, they comprise all the system's costs, revenue and financial parameters.

### Locations and weather analysis

Among many key requirements for selecting the best sites for installing solar power plants, the availability of good solar resources is the most important factor (Meyer et al., 2012). As the diffuse irradiance cannot be optically concentrated, the solar resource of CSP systems only consists of the direct normal irradiance (DNI), which is by far the important parameter for energy yield calculation and performance assessment of CSP systems (Janotte et al., 2017). The DNI corresponds to the amount of solar irradiance that is perpendicular to the unit area received from the incoming rays from the sun. Typically, for a CSP system to achieve high economic performance, the range of cumulative annual DNI at a site must vary between 1600 and 2000 kWh/m<sup>2</sup>/year (Parrado et al., 2016; Wang, 2019). Being close to the equator, Rwanda receives abundant sunshine throughout the whole year. The global solar radiation about 4.8–5.2 kWh/m<sup>2</sup>/day and peak sun hours of approximately 8 h per day indicate the great potential of solar technologies in the country (Museruka & Mutabazi, 2007; Safari & Gasore, 2009).

Figure 2 shows the DNI map of Rwanda, which was obtained from Solargis global solar model developed by the World Bank Group (Solargis, 2019). As it can be seen in Fig. 2, the DNI is the highest in the Southern and Eastern

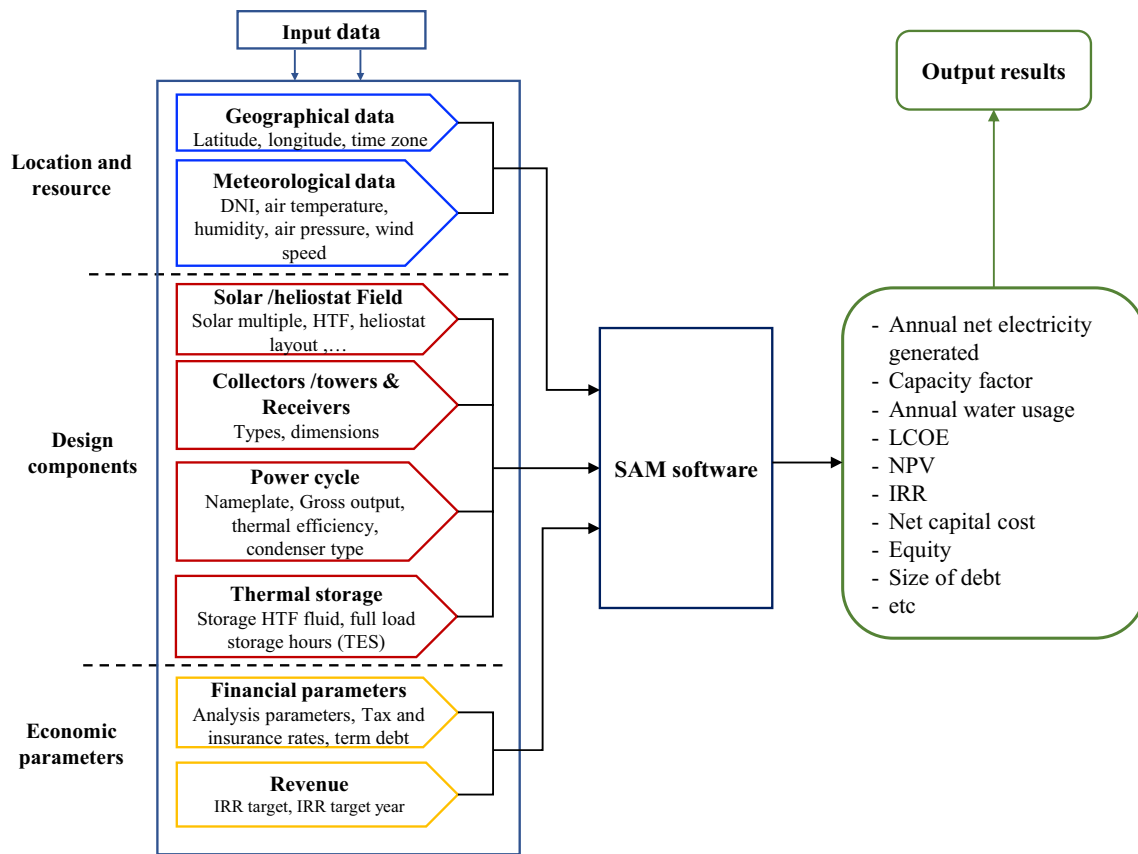


Fig. 1 Flowchart of the simulation in SAM software (adapted from (Trabelsi et al., 2016))

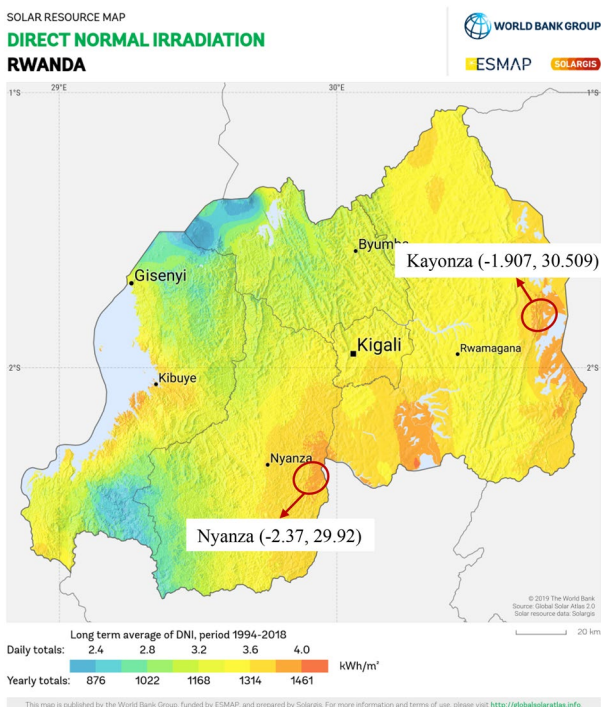


Fig. 2 DNI distribution map in Rwanda

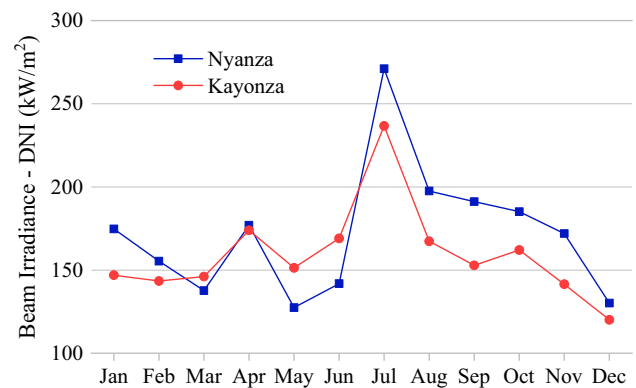


Fig. 3 Monthly average of DNI for the two selected locations

parts of Rwanda. For this reason, two locations were selected for this study, namely Nyanza, located on latitude  $-2.37^{\circ}\text{E}$  and longitude  $29.92^{\circ}\text{S}$ ; and Kayonza, located on latitude  $-1.907^{\circ}\text{E}$  and longitude  $30.509^{\circ}\text{S}$ .

The DNI values for these two locations were retrieved from the typical meteorological year (TMY) data by the European Commission PVGIS and were recorded for a period from 2005 to 2016 (European Commission, 2019). The average monthly solar radiations for these locations are

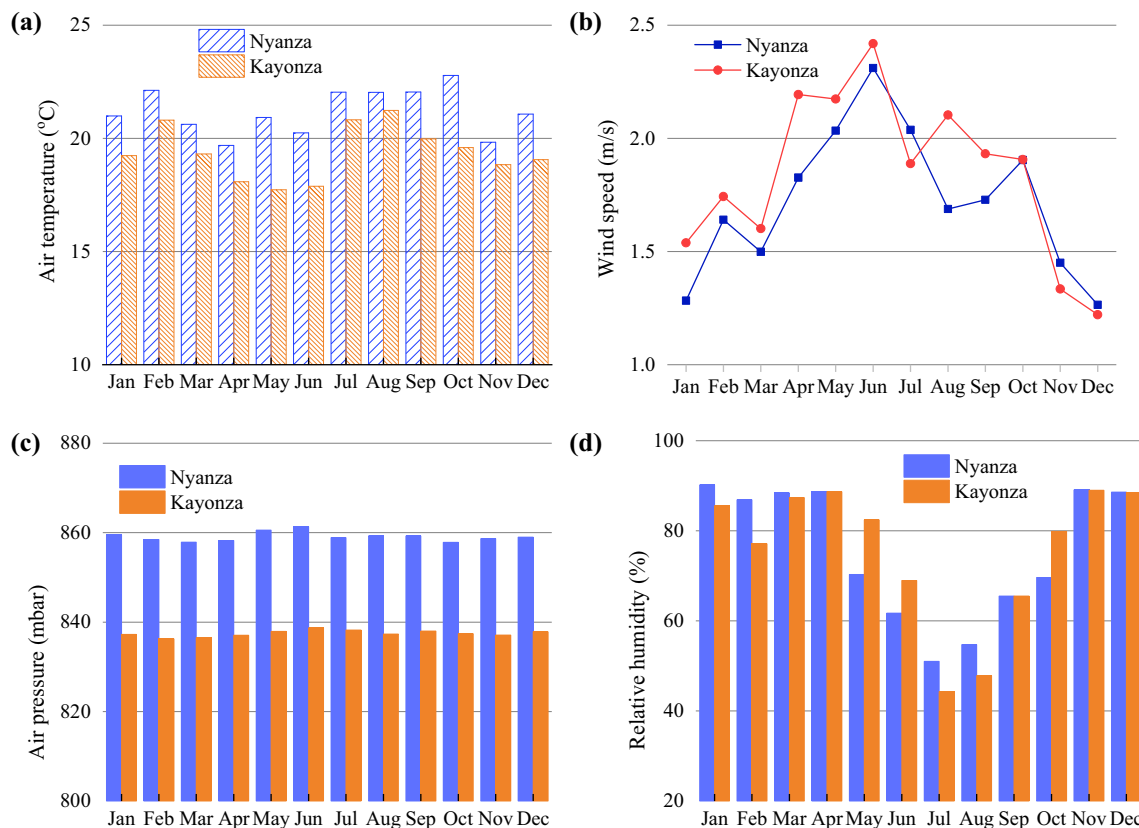
shown in Fig. 3. The DNI reach their maximum values in July for both locations, corresponding to the dry season in Rwanda; while minimal in May for Nyanza and in December for Kayonza, as a result of the rainy seasons within those periods.

The operation and performance of solar power plants also require the knowledge of meteorological conditions of the selected location. The increase in ambient temperature and relative humidity reduce significantly the efficiency of the CSP plant, hence resulting in a lower electricity production (Kumar et al., 2021); wind may also lift dust and other contaminants which cause the degradation of the optical efficiency of collectors and therefore affect the performance of the CSP plant (Sultan et al., 2020). For modeling CSP plants, meteorological data of these parameters are usually used as input parameters in the SAM software. Figure 4 shows the TMY data in Nyanza and Kayonza, including air temperature, wind speed, relative humidity and air pressure (European Commission, 2019). According to Fig. 4a, the maximum temperature values for Nyanza and Kayonza are recorded in October and August, respectively, while minimum temperature values are recorded in April and May, respectively. Meanwhile, as shown in Fig. 4b, the wind speed for both locations reaches its maximum value in June, and it

is minimal in December. The air pressure (shown in Fig. 4c) is almost uniform during the whole year, slightly higher in Nyanza than in Kayonza; higher values can be obtained in June for both locations, while lower values are obtained in October for Nyanza and in February for Kayonza. As for the relative humidity (Fig. 4d), highest values are recorded in November for Kayonza and in January for Nyanza, while lowest values are found in July for both locations.

### Modeling the performance of PTPP and STPP

To model the performance of both CSP technologies, various technical assumptions were considered including the specifications of HTF for the solar field; configuration of power cycle and thermal storage parameters; configuration of collectors and receiver components for the PTPP; and optimization of heliostat field and towers for STPP. The modeling considered the most commercially available components which have been used in operational CSP installations. For the economic assumptions, different financial parameters such as project term debts, financial rates and taxes were updated according to the economic indicators



**Fig. 4** Monthly average meteorological data for Nyanza and Kayonza, Rwanda. **a** Air temperature **b** Wind speed **c** Air pressure **d** Relative humidity



of Rwanda. The following sub-sections provide details of both technical and economic assumptions used in this study.

### Technical parameters

The technical assumptions used for the modeling of the PTPP presented in this paper are summarized in Table 1. For the solar field, this study uses Therminol VP-1 as heat transfer fluid (HTF), a mixture of biphenyl and diphenyl oxide, with a design loop inlet temperature of 293 °C and design loop outlet temperature of 391 °C. The Therminol VP-1 has been widely used in CSP for cost analysis owing to its low freezing point and low viscosity (Eastman Chemical Company 2021; Hirbodi et al., 2020). This study also uses the SkyTrough™ collector (developed by SkyFuel), with 8 modules per solar collector assembly (SCA) and 4 SCA are connected in series to form a loop. The SCA has a length of 115 m, an aperture width of 6 m, and each loop has a reflective aperture area of 656 m<sup>2</sup>. This type of collector has been widely researched for cost analysis and has been proven to reduce capital cost, shorten installation time, and reduce operation and maintenance costs (Awan et al., 2019; Farr & Gee, 2009). Meanwhile, the Schott PR80 was used as receiver, which has 7 cm absorber tube inner diameter and 12 cm glass envelope outer diameter. For the power cycle, a steam Rankine Cycle was opted for in this study, with cycle thermal

efficiency of 39% (Hirbodi et al., 2020); a condenser type with dry cooling technology was also selected for the modeling, by considering the unavailability of water resources in the selected sites (Agyekum & Velkin, 2020; Aly et al., 2019). As for the TES system, the HITEC solar molten salt was used as a storage HTF, with 250 °C cold tank heater set point and 365 °C hot tank heater set point. Molten salts are the most widespread storage materials in CSP plants owing to their low cost and considerable properties such as high thermal stability, low viscosity and non-toxicity (Tian & Zhao, 2013).

The technical assumptions for the modeling of STPP are also illustrated in Table 2. The design point DNI was fixed at 950 W/m<sup>2</sup> and the cycle thermal efficiency at 40% (Kincaid et al., 2018). Molten salt (60% NaNO<sub>3</sub> + 40% KNO<sub>3</sub>) was selected as the HTF with HTF hot and cold temperature of 574 and 290 °C, respectively. This type of salt is widely used in STPP due to its stability at temperatures as high as 600 °C (Vignarooban et al., 2015). Molten salt not only allows the solar field to operate at higher temperatures but also it is less expensive and more environmental friendly than other HTF (Zhang et al., 2018). Meanwhile, unlike for the case of PTPP, the design of STPP also requires the optimization of heliostat field layout and tower dimensions to match the receiver thermal power at the design point. Therefore, the heliostat field was always optimized for each solar field size and the values of the

**Table 1** Technical input parameters used for the design of the PTPP model

| Component                 | Parameter                      | Description               |
|---------------------------|--------------------------------|---------------------------|
| Solar field               | Heat transfer fluid (HTF)      | Therminol VP-1            |
|                           | HTF min operating temperature  | 12 °C                     |
|                           | HTF max operating temperature  | 400 °C                    |
|                           | Loop inlet temperature         | 293 °C                    |
|                           | Loop outlet temperature        | 391 °C                    |
| Solar collectors assembly | Collector type                 | SkyFuel SkyTrough         |
|                           | Number of modules per assembly | 8                         |
|                           | Reflective aperture area       | 656 m <sup>2</sup>        |
|                           | Length of collector assembly   | 115 m                     |
|                           | Mirror reflectance             | 0.93                      |
| Receivers                 | Receiver type                  | Schott PTR80              |
|                           | Absorber tube inner diameter   | 0.076 m                   |
|                           | Glass envelop outer diameter   | 0.12 m                    |
| Power cycle               | Design gross output            | 111 MWe                   |
|                           | Nameplate                      | 100 MWe                   |
|                           | Cycle thermal efficiency       | 0.39                      |
|                           | Turbine inlet pressure control | Fixed pressure            |
|                           | Condenser type                 | Air cooled                |
| Thermal storage (TES)     | Storage HTF fluid              | Hitec Solar Salt          |
|                           | Cold tank heater set point     | 250 °C                    |
|                           | Hot tank heater set point      | 365 °C                    |
|                           | HTF density                    | 1872.49 kg/m <sup>3</sup> |

**Table 2** Technical input parameters used for the design of the STPP model

| Component             | Parameter                       | Description                                                 |
|-----------------------|---------------------------------|-------------------------------------------------------------|
| System design         | Design point DNI                | 950 W/m <sup>2</sup>                                        |
|                       | HTF hot temperature             | 574 °C                                                      |
|                       | HTF cold temperature            | 290 °C                                                      |
|                       | Design turbine gross output     | 111MWe                                                      |
|                       | Cycle thermal efficiency        | 0.4                                                         |
| Heliostat field       | Heliostat layout and dimensions | Always optimize                                             |
|                       | Heliostat width                 | 12.2 m                                                      |
|                       | Heliostat height                | 12.2 m                                                      |
|                       | Number of heliostat facets—X    | 2                                                           |
|                       | Number of heliostat facets—Y    | 10                                                          |
|                       | Heliostat focusing method       | Ideal                                                       |
| Tower and receiver    | Heliostat canting method        | On-axis                                                     |
|                       | Tower height                    | Obtained from optimization                                  |
|                       | Receiver height                 | Obtained from optimization                                  |
|                       | Receiver diameter               | Obtained from optimization                                  |
|                       | HTF type                        | Molten salt (60% NaNO <sub>3</sub> + 40% KNO <sub>3</sub> ) |
| Power cycle           | Flow pattern                    | Configuration 1                                             |
|                       | Nameplate                       | 100 MW                                                      |
|                       | Condenser type                  | Air cooled                                                  |
|                       | Ambient temperature at design   | 42 °C                                                       |
| Thermal storage (TES) | Storage type                    | Two tank                                                    |
|                       | Tank height                     | 20 m                                                        |

**Table 3** Financial input parameters for the PTPP and STPP models

| Financial parameters            | Value                           | References                              |
|---------------------------------|---------------------------------|-----------------------------------------|
| Analysis parameters             |                                 |                                         |
| Analysis period                 | 25 years                        |                                         |
| Inflation rate                  | 2.7% per year                   | (Trading Economics, 2021a)              |
| Real discount rate              | 12% per year                    | (Hernández-Moro & Martínez-Duart, 2013) |
| Project term debt               |                                 |                                         |
| Debt percent                    | 70% of the total installed cost |                                         |
| Tenor                           | 18 years                        |                                         |
| Moratorium                      | 5 years                         |                                         |
| Annual interest rate            | 4.50%                           | (Trading Economics, 2021b)              |
| Up-front fee                    | 2.75% of the total debt         |                                         |
| Project tax and insurance rates |                                 |                                         |
| Sales tax                       | 18%                             | (Trading Economics, 2021c)              |
| Corporate income tax rate       | 30%                             | (RRA, 2021; Trading Economics, 2021d)   |
| Annual insurance rate           | 0.5% of the installed cost      |                                         |
| Construction period debt        |                                 |                                         |
| Up-front fee                    | 1% of the principal             |                                         |
| Months prior to operation       | 24 months                       |                                         |
| IRR target                      | 15%                             | (Aly et al., 2019)                      |
| IRR target year                 | 20                              |                                         |

tower height, receiver height and receiver diameter were obtained from optimization.

### Financial parameters

The financial assumptions used as input values for the modeling of PTTP and STTP are summarized in Table 3. The analysis period of the project was selected at 25 years. Various economic indicators of Rwanda were obtained from the Trading Economics website, including the inflation rate of 2.7% (Trading Economics, 2021a), annual interest rate of 4.5% (Trading Economics, 2021b) and sales tax of 18% (Trading Economics, 2021c). The corporate tax rate of 30% was obtained from the Rwanda Revenue Authority website (RRA, 2021). The annual discount rate and IRR target were also selected as 12% and 15%, respectively, with respect to the recommended values (Aly et al., 2019; Hernández-Moro & Martínez-Duart, 2013). Defaults values from SAM software were otherwise used.

### Technical and economic performance indicators

To assess the technical and economic performance of CSP power plants, various technical indicators need to be defined. The key technical indicators adopted in this study include:

- *The annual net electricity generated (AEG)* of the plant which is simply the difference between the gross power output ( $E_{gross}$ ) and the power output from the parasitic losses ( $E_{par}$ ). It is shown that the AEG can be calculated by the following equation (Awan et al., 2019; Desideri & Campana, 2014):

$$AEG = \sum_{h=1}^{8760} (Q_{ST} A_{sf} \eta_{sh} \eta_{soil} \eta_{sto} \eta_{PB} \eta_{ta} - E_{par}) \quad (1)$$

where  $Q_{ST}$  is the hourly thermal energy output per unit square meter of collector;  $A_{sf}$  is the solar field area;  $\eta_{sh}$  is the shading efficiency;  $\eta_{soil}$  is the soiling effect;  $\eta_{sto}$  is the thermal storage efficiency;  $\eta_{PB}$  is the power block efficiency; and  $\eta_{ta}$  is the turbine-alternator set efficiency.

- *The capacity factor (CF)* which is used to indicate the ratio between actual generation capacity of the plant and the power plant's maximum possible generation capacity. The CF is given by the equation below:

$$CF = \frac{AEG}{8760 \cdot E_{gross}} \quad (2)$$

- *Annual water usage* which is the amount of water used for washing the components, steam generation and cooling of the plant in a period of one year.

Various economic indicators are also important in assessing the economic performance of a CSP power plant. The key economic indicators used in this study are described as follows:

*The Levelized cost of electricity (LCOE)* is one of the most commonly used metric for the economic performance of RE projects. The LCOE is expressed as the sum of all the costs incurring during the lifetime of a project divided by the units of the energy produced during its lifetime. A lower LCOE implies a more economically viable project. In SAM software, both nominal and real LCOE are calculated. Since the nominal LCOE is the current dollar value and accounts for short-term projects, in this study we only consider the real LCOE which is the constant dollar value and accounts for the long years of inflation over the project lifetime. The LCOE of a solar plant can be calculated using the equation below (IRENA, 2012; Sultan et al., 2020):

$$LCOE = \frac{\sum_{t=1}^n \left[ \frac{I_t + OM_t + F_t}{(1+d)^t} \right]}{\sum_{t=1}^n \left[ \frac{E_t}{(1+d)^t} \right]} \quad (3)$$

where  $I_t$  is the investment expenditures in the year  $t$ ;  $OM_t$  represents the operation and maintenance expenditures in the year  $t$ ;  $F_t$  represents the fuel expenditures in the year  $t$ ;  $E_t$  is the electricity generation in the year  $t$ ;  $d$  is the discount rate; and  $n$  is the life of the plant.

*The Net Present Value (NPV)* is another important economic indicator used in assessing the economic viability of solar plants. The NPV of a project is a measure of its profitability taking into account both revenue and cost of an investment. It is basically the present value of the after tax cash flow discounted on an average rate of discount rate in excess of the present value of the investment. The NPV can be calculated according to equation below. A positive NPV means the project is commercially viable, while a negative NPV indicates otherwise.

$$NPV = -C + \sum_{t=1}^T \frac{Q_t}{(1+d)^t} \quad (4)$$

In this Eq. (4),  $Q_t$  represents the net cash flow in year  $t$  and is calculated using the equation below:

$$\begin{aligned} Q_t &= (\text{cashinput})_t - (\text{cashoutput})_t \\ &= (E_{tariff} E_{gen}) - O\&M \end{aligned} \quad (5)$$

where  $E_{tariff}$  is the electricity tariff for the CSP system,  $E_{gen}$  is the annual energy generation.

*The internal rate of return (IRR)* is also another key economic performance indicator to assess a project's profitability. The IRR is a discount rate at which the NPV is equal to zero and is given using the following equation:



$$NPV = -C + \sum_{t=1}^T \frac{Q_t}{(1 + IRR)^t} = 0 \quad (6)$$

There are two relevant IRRs, namely the project's IRR and the equity IRR. The project IRR gives the rate of return at the end of the whole project, without considering the debt in the project financing; while the equity IRR only considers the cash flow for the equity shareholders such as after-tax revenue, operating expenses and debt payments.

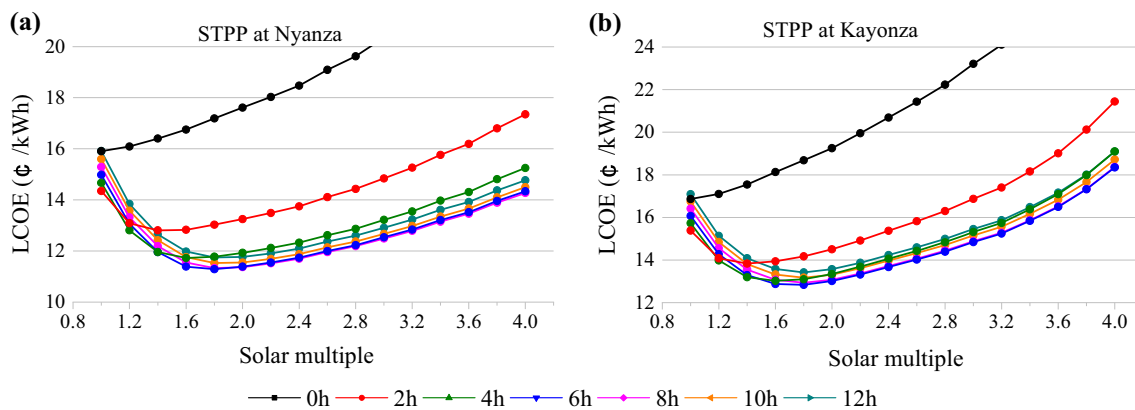
## Results and discussion

This section presents the results of the optimized 100 MW PTPP and STPP at the selected two locations, Nyanza and Kayonza. We assess the performance of the plants by investigating the variation of key technical indicators such as the AEG, annual water usage and CF, as well as key economic indicators such the LCOE, NPV and IRR.

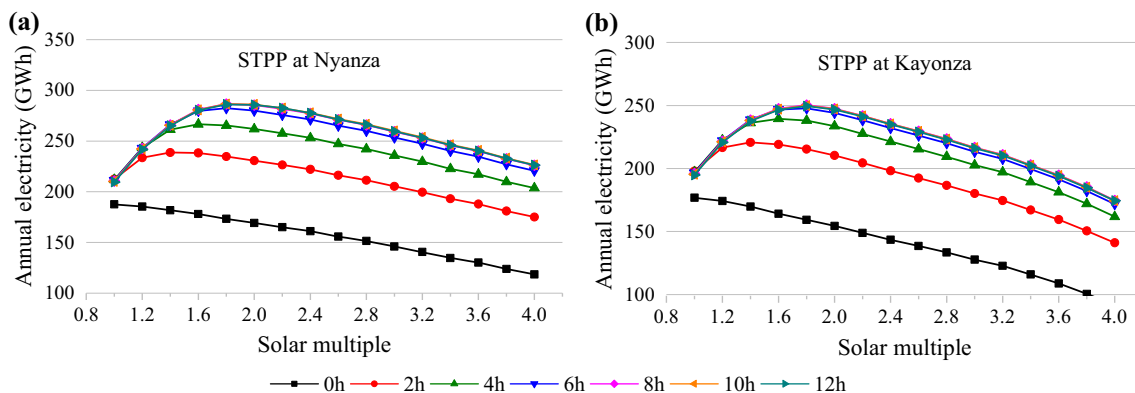
These indicators are then used for a sensitivity analysis to address the uncertainty of input data and to identify parameters which have the biggest impact on the overall performance of the power plant.

### Optimization of the STPP for both sites

Figures 5 and 6 show the variation of LCOE and AEG with respect to the SM for the modelled STPP at Nyanza and Kayonza under different TES capacities. Based on Fig. 5, it can be seen that for an STPP system without the TES, an increase in the SM results in a significant increase of the LCOE. For the system integrated with the TES, the LCOE at any given TES first decreases with increasing SM until it reaches a minimum value then starts rising again. These results are consistent with those reported in previous studies (Aly et al., 2019; Hirbodi et al., 2020; Qoaidar & Liqreina, 2015). Figure 5 also shows that for any given TES size, the optimum SM which gives the minimum LCOE falls within the region of 1.4–2.0. According to the figure, the lowest



**Fig. 5** Variation of LCOE with SM for a 100 MW STPP modelled in Nyanza site (a) and Kayonza site (b) at different TES capacities (full load hours 0–12 h)



**Fig. 6** Variation of AEG with SM for a 100 MW STPP modelled in Nyanza site a and Kayonza site b at different TES capacities

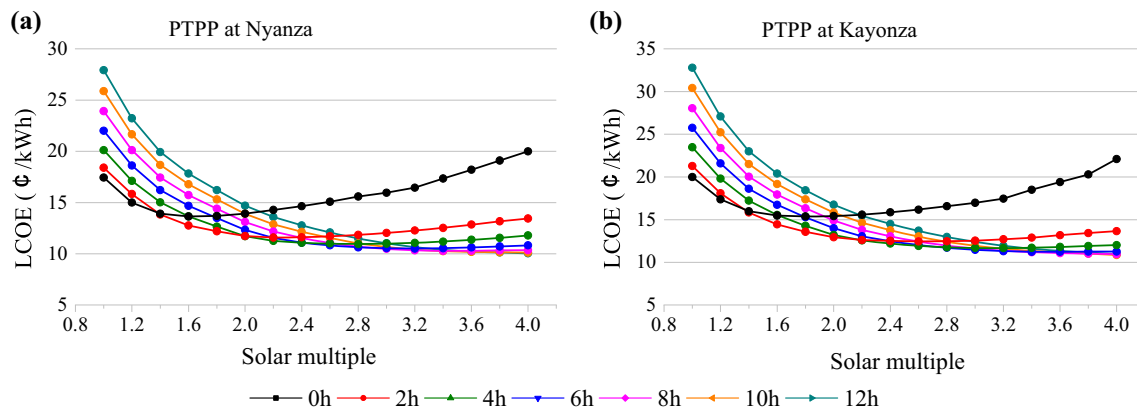
value of LCOE was obtained at  $SM = 1.8$ , which clearly indicates that the  $SM = 1.8$  is the optimal solar field size for the STPP at both sites. Moreover, the minimum LCOE is decreasing slightly with increasing the TES capacity from 0 to 8 h, but then increases again for TES sizes larger than 8 h.

Meanwhile, as Fig. 6 shows, the amount of annual electricity generated for any given TES is rising with the increase of SM until it reaches a certain maximum then starts decreasing again. Larger solar field generally lead to lower performance of the plant due to the poor optical and thermal performance of solar components, thus lowering the electricity generated. Similar trends have been reported in previous studies (Boudaoud et al., 2015; Hirbodi et al., 2020). Moreover, the figure also shows that when the TES capacity is increased, the annual electricity generated is also increased until the 8 h period of TES. A further increase of TES doesn't affect the AEG anymore. Therefore, it is clear that the 8 h of full load storage is the optimal TES capacity for the STPP at both sites as it gives the highest amount of annual electricity generated.

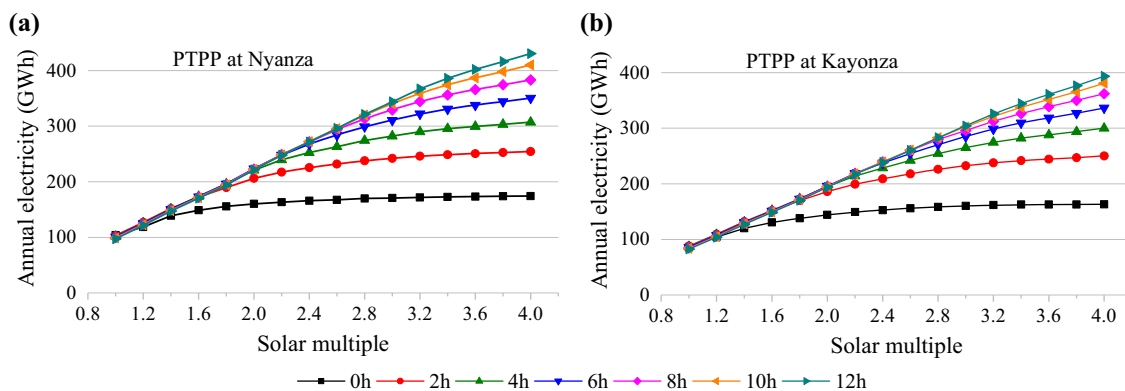
Accordingly, the optimized STPP configuration at both sites was then obtained at  $SM = 1.8$  and  $TES = 8$  h, corresponding to the combination of both SM and TES which give the lowest LCOE and the highest AEG.

### Optimization of the PTPP for both sites

Figures 7 and 8 show the variation of LCOE and AEG with respect to the SM for the modelled PTPP at Nyanza and Kayonza under different TES capacities. It is clear from Fig. 7 that the trend of the LCOE for the PTPP is different than the STPP. Regardless whether the system is integrated with or without the TES, the LCOE always tends to first decrease with an increase in SM, reaches its minimum and then increases again. It is observed from the results that for a system without the TES, the optimum SM is 1.5, while for a system integrated with TES, the optimum SM depends largely on the TES size. When the TES is lower than 6 h, the lowest LCOE is obtained at  $SM = 2.2, 2.8$  and  $3.2$  for  $TES = 2$  h, 4 h and 6 h, respectively. However, for TES larger



**Fig. 7** Variation of LCOE with SM for a 100 MW PTPP modelled in Nyanza site (a) and Kayonza site (b) at different TES capacities



**Fig. 8** Variation of AEG with SM for a 100 MW PTPP modelled in Nyanza site (a) and Kayonza site (b) at different TES capacities

than 6 h, the LCOE keeps decreasing within the initial SM range and doesn't have any minimum, which suggests that the selected SM for this study is lower than the expected SM for this specific TES capacity. This trend of LCOE for PTPP echoes the one reported in a previous study (Agyekum & Velkin, 2020).

The similar effect can also be observed by the variation of the AEG with respect to SM, as shown in Fig. 8. For any given TES size lower than 6 h, the AEG increases with the increase of the SM until the SM = 3.2, then remains almost constant. For TES sizes larger than 6 h, the AEG values keep increasing, clearly because the selected SM range in this study is lower than the expected SM for higher TES capacity.

Meanwhile, although the further increase in SM may decrease the LCOE (conversely increase the amount of AEG), but it also implies a significantly higher cost of the plants. Table 4 illustrates the minimum LCOE values at each TES size, and the corresponding SM and the net capital cost (NCC). In this study, the configuration where the minimum LCOE decreased to less than 5% compared to the minimum

LCOE of the previous TES size was selected as the optimal PTPP design. This approach has been employed in a previous study (Aly et al., 2019), taking into account that when choosing the appropriate design, CSP developers always tend to consider the configuration which gives a balance between minimizing the LCOE and increasing the project's net capital cost. Therefore, based on our results, the optimal PTPP design at both sites was selected at SM = 3.2 and TES = 6 h.

### Performance analysis of the optimized plants

The technical and economic results from the simulation of the optimized plants (i.e. SM = 1.8, TES = 8 h for STPP and SM = 3.2, TES = 6 h for PTPP) are presented in Table 5. For the optimized STPP, an AEG of about 286 GWh and 250 GWh was obtained at Nyanza and Kayonza, respectively, with corresponding CF of 32.7% and 28.6%; while the optimized PTPP achieved an AEG of about 322 GWh and 298 GWh at Nyanza and Kayonza, respectively, with

**Table 4** Optimal SM at each TES size and their corresponding minimum LCOE and NCC in Nyanza and Kayonza

| TES (h) | Nyanza     |                   |                   |                  | Kayonza    |                   |                   |                  |
|---------|------------|-------------------|-------------------|------------------|------------|-------------------|-------------------|------------------|
|         | Optimal SM | Min. LCOE (¢/kWh) | LCOE decrease (%) | NCC (million \$) | Optimal SM | Min. LCOE (¢/kWh) | LCOE decrease (%) | NCC (million \$) |
| 0       | 1.6        | 13.64             | –                 | 411              | 1.8        | 15.39             | –                 | 442              |
| 2       | 2.2        | 11.59             | 15.03             | 556              | 2.6        | 12.45             | 19.10             | 615              |
| 4       | 2.8        | 10.95             | 5.52              | 698              | 3          | 11.66             | 6.35              | 728              |
| 6       | 3.2        | 10.51             | 4.02              | 811              | 3.4        | 11.24             | 3.60              | 841              |
| 8       | 3.4        | 10.28             | 2.19              | 893              | 3.8        | 10.95             | 2.58              | 986              |
| 10      | 4          | 10.12             | 1.56              | 1037             | 4          | 10.86             | 0.82              | 1037             |
| 12      | 4          | 10.04             | 0.79              | 1089             | 4          | 10.84             | 0.18              | 1089             |

**Table 5** Technical and economic results of the optimized PTPP and STPP configurations for Nyanza and Kayonza

| Metric                               | STPP (SM = 1.8, TES = 8 h) |             | PTPP (SM = 3.2, TES = 6 h) |             |
|--------------------------------------|----------------------------|-------------|----------------------------|-------------|
|                                      | Nyanza                     | Kayonza     | Nyanza                     | Kayonza     |
| AEG, kWh                             | 286,578,048                | 250,284,832 | 322,219,296                | 298,760,448 |
| Capacity factor, %                   | 32.7                       | 28.6        | 36.8                       | 34.1        |
| Annual water usage, m <sup>3</sup>   | 77,604                     | 74,827      | 91,665                     | 89,285      |
| PPA price (year 1), ¢/kWh            | 11.86                      | 13.57       | 10.07                      | 10.78       |
| Levelized PPA price (nominal), ¢/kWh | 14.81                      | 16.90       | 13.74                      | 14.78       |
| Levelized PPA price (real), ¢/kWh    | 11.57                      | 13.20       | 10.73                      | 11.54       |
| LCOE (nominal), ¢/kWh                | 14.50                      | 16.54       | 13.45                      | 14.47       |
| LCOE (real), ¢/kWh                   | 11.33                      | 12.92       | 10.51                      | 11.30       |
| NPV, \$                              | 5,441,854                  | 5,444,884   | 5,681,750                  | 5,683,988   |
| IRR, %                               | 30.46                      | 30.45       | 30.46                      | 30.45       |
| IRR at end of project, %             | 30.64                      | 30.63       | 30.64                      | 30.63       |
| Net capital cost \$                  | 777,324,672                | 777,258,880 | 811,472,448                | 811,423,808 |
| Equity, \$                           | 233,395,152                | 233,375,392 | 243,648,176                | 243,633,568 |
| Size of debt, \$                     | 543,929,536                | 543,883,456 | 567,824,256                | 567,790,272 |

corresponding CF of 36.8% and 34.1%. The modelled real LCOEs are 11.33 ¢/kWh and 12.92 ¢/kWh for the optimized STPP in Nyanza and Kayonza, respectively; while they are 10.51 ¢/kWh and 11.30 ¢/kWh for the optimized PTPP in Nyanza and Kayonza, respectively. These obtained LCOE values are obviously lower than the global weighted average LCOE for CSP plants which was estimated at 18.2 ¢/kWh in 2019 (IRENA, 2020). Furthermore, the positive NPV also suggests that both plants are economically viable in these sites. Meanwhile, the lowest LCOE and the highest AEG are obtained at Nyanza, suggesting that Nyanza is the best location for the installation of both technologies.

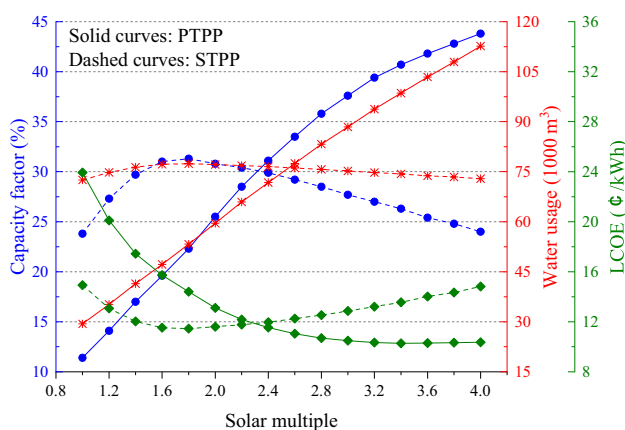
### Sensitivity analysis

To assess the impact of key technical and economic parameter assumptions on the efficiency and profitability of the CSP plants at both sites, a sensitivity analysis is carried out. By doing so, input parameters can be chosen carefully depending on whether they have a significant impact on the overall performance of the plant. Usually, a sensitivity analysis is performed by varying one parameter while others are kept constant. As the parameters have the same impact and present results with similar trends, in this section only the results of sensitivity analysis for PTPP and STPP at Nyanza site are presented, and the plant design was fixed at TES = 8 h and SM = 1.8, which corresponds to the optimized STPP configuration. Herein, the effect of SM and TES size on the technical and economic performance was assessed, followed by the effect of annual interest rate and discount rate on the economic performance of the plants.

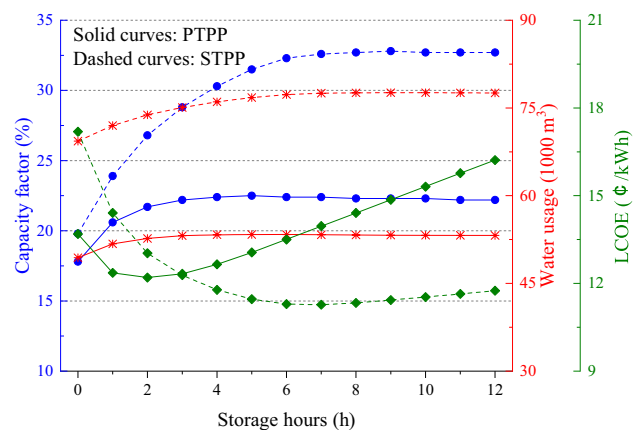
Figure 9 shows the sensitivity of CF, annual water usage and LCOE to the variation of SM for both PTPP and STPP. Generally having a higher SM in CSP plants implies having a larger solar field to collect solar radiation. As shown in the figure, when the SM is increased, the CF of the STPP

initially increases sharply until it reaches its maximum, corresponding to the highest intensity of electricity generation at the optimal technical performance of the plant. As a result, the LCOE also decreases since it largely depends on the cost and electricity generation. When the SM is further increased, the CF decreases, that is, the intensity of electricity generation reduces, and only the operation and maintenance costs are increased; as a result the LCOE increases (Suresh et al., 2014). It can also be observed from Fig. 9 that the increase in SM has little effect on the variation of the annual water usage for the STPP. As for the PTPP, the CF keeps increasing sharply with the increase of the SM, while the LCOE for the PTPP decreases sharply at smaller SM, then slightly at higher SM. This clearly implies that the selected solar field range is still not enough to collect sufficient solar energy for operating the plant (Hirbodi et al., 2020). It is also clear that the annual water usage for the PTPP has an increasing trend with the solar field size, which is expected as the water consumption is mainly used for cleaning the solar field components (Boudaoud et al., 2015). Meanwhile, the figure also indicates that, at smaller SM, the STPP have a higher CF and lower LCOE than PTPP, implying that the STPP produce more electricity and are suitable at smaller SM than PTPP; conversely the PTPP have smaller LCOE and higher CF at higher SM.

Figure 10 shows the effect of the variation of TES on the technical and economic performance of the power plants. It can be seen from the figure that the effect of TES period to the CF is more significant for the STPP than the PTPP. The CF increases slightly until 4 h of TES for the PTPP and 8 h of TES for the STPP, then remains almost constant after those periods. This means that the thermal energy collected by the solar field is high enough for the plants to generate electricity at their optimal capacity. The extra thermal energy collected by the solar field is then stored by the TES system for generating electricity after sunset.



**Fig. 9** Sensitivity analysis of CF, water usage and LCOE to the variation of solar multiple (at TES = 8 h)



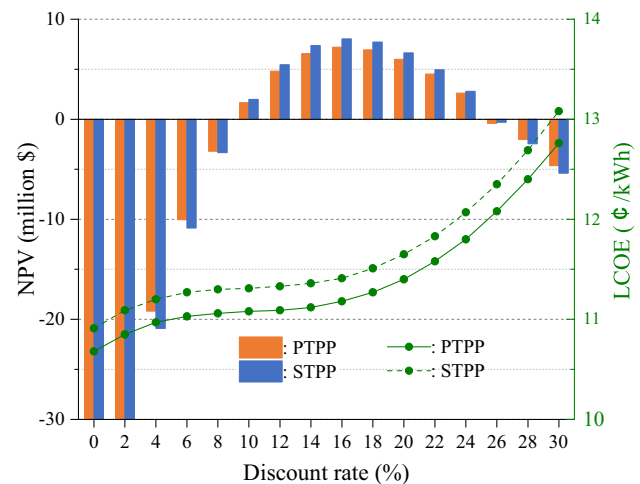
**Fig. 10** Sensitivity analysis of CF, water usage and LCOE to the variation of TES size (at SM = 1.8)

Further increase of TES size does not affect significantly the CF as all the excess thermal energy has been stored in the TES system. It is also clear from Fig. 10 that the TES size has a minor effect on the annual water usage for both technologies as the curves are almost flattened. Moreover, it can be observed that the LCOE for both plants decreased until the corresponding CF reached its maximum, then increased significantly for the PTPP while it remained almost constant for the STPP. The figure also shows that the variation curves of LCOE for PTPP and STPP intersect each other at a specific TES capacity (at about 3 h period of TES). Before the intersection point, the LCOE values for the STPP are higher than the PTPP, then become lower after the intersection point. Such behaviors were also obtained in previous studies (Hirbodi et al., 2020). This result implies that the STPP are more suitable at higher TES sizes and may explain why most commercial STPP are integrated with relatively high TES capacities, such as the 20 MW Gemasolar Thermosolar Plant in Spain with 15 h TES and the 100 MW Redstone Solar Thermal Power Plant in South Africa with 12 h TES (NREL, 2021).

We have also carried out a sensitivity analysis of the economic performance to the variation of the annual interest rate and discount rate to indicate the large influence these parameters have on the economic feasibility of CSP projects. In certain cases depending on the kind of project, private energy developers may face higher debt interest rates than the government-led power generation projects (Aly et al., 2019). Therefore, it is deemed imperative to address the financial uncertainty caused by the changing conditions of the project term debt. Figure 11 shows that the annual interest rate has a significant impact on the economic feasibility of both the CSP plants. Increasing the interest rate results in an improvement of NPV, but also it causes a significant increase in LCOE and decrease of IRR. For instance, for the case of STPP, the LCOE would increase from 11.33 ¢/kWh

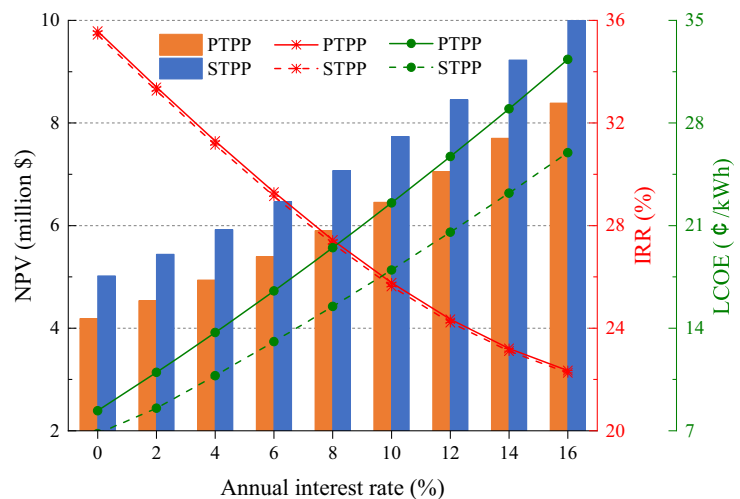
to 20.55 ¢/kWh when the interest rate is increased from 4.5% (base case from central Bank (Trading Economics, 2021b)) to 12% (private sector-led projects (African Development Bank, 2021)), while the IRR would reduce from 30.64 to 24.22% in the same range.

Another key input parameter is the discount rate which acts as a measure of time value of the money and is central to the calculation of the present value. Discount rate also takes into account the risk of the investment and thus the choice of a discount rate is vital to any economic analysis. It is found that the discount rates of CSP systems are high, usually between 10 and 15%, as a result of high technological risks associated with CSP projects (Hernández-Moro & Martínez-Duart, 2013; Zhao et al., 2017). Therefore, it is very important to assess carefully the basis for choosing the value of discount rate as it affects the overall economic feasibility of the CSP projects and therefore can influence



**Fig. 12** Sensitivity analysis of the NPV and LCOE to the variation of discount rate

**Fig. 11** Sensitivity analysis of the NPV, IRR and LCOE to the variation of annual interest rate





investors' decisions towards one option or another in undertaking the project (NREL, 2010). Results from the Fig. 12 indicate that the discount rate has a huge impact on the NPV while it has a slight increasing effect on the LCOE. It can be observed that both technologies are only economically competitive when the discount rate is between 10 and 24% since the NPV is positive in that range. Meanwhile, the variation of LCOE in the positive NPV range is fairly minimal and varies between 11.31 ¢/kWh (at 10% discount rate) and 12.07 ¢/kWh (at 24% discount rate) for STPP, and between 11.03 ¢/kWh and 11.80 ¢/kWh in the same range for the PTPP. In this study, a discount rate of 12% was then selected as the base case as it gives a positive NPV at a relatively low LCOE.

### Major barriers and policy implications

In this section, we discuss the major reasons why RE technologies such as CSP are not deployed in Rwanda. Several barriers that hamper the deployment of CSP technologies in Rwanda can be categorized into the financial barriers; technical barriers; and policy and regulatory barriers.

#### Financial barriers

The high cost of capital associated with the CSP technologies constitutes a significant barrier for Rwanda. As a land-locked country, all the necessary RE equipment are sourced and imported from abroad which increases the upfront cost of setting up a RE power plant (Mukeshimana et al., 2021b). Some other financial barriers include the lack of private sector participation and the reluctance of local financial institutions to finance RE projects, mostly due to risk perceptions associated with such projects (Labordena et al., 2017; Mukeshimana et al., 2021a).

Efforts are needed on the government side for allocating financial investment through several international funds and financial instruments to reduce the cost of financing the projects (Michaelowa et al., 2021). Various incentives and policy supports are also needed to attract private sector and to encourage entrepreneurs to step in and invest in solar energy technologies, such as tax exemptions on power equipment and provision of land for power projects, road access and water supply (Ouedraogo, 2019; RDB, 2020). Some other measures can be adopted as well such as implementing the feed-in mechanism; and applying creative and effective financing schemes with relatively longer pay-back periods (Mukeshimana et al., 2021a).

#### Technical barriers

The lack of accurate solar resources data is one of the main reasons that hinder the growth of Rwandan solar energy market.

CSP requires appropriate knowledge of DNI data, the lack of which makes it almost impossible to predict the overall techno-economic feasibility of the CSP installation (Kiefer & del Río, 2020). Moreover, the lack of R&D activities in RE sector, the lack of skilled personnel and training in energy project and management, also makes it very challenging to install RE power plant in Rwanda (Mukeshimana et al., 2021b).

It is required to set up the needed network of high-quality meteorological measurement facilities. With the availability of accurate solar resources data, it would be feasible to obtain appropriate techno-economic assessment of the CSP plants, which would then reduce risk and uncertainty. Elsewhere, investing in R&D activities in CSP sector and establishing institutions which can manufacture CSP components will also be key to increase the efficiency and to reduce the installation cost of CSP systems. Meanwhile, long-term plans and training programs related to CSP technologies may also be initiated to develop a strong workforce and to build capacity throughout the energy value chain.

#### Policy and regulatory barriers

The success of CSP not only depends on financial aspects but also requires some level of government support and policy settings. CSP is a highly-capital intensive technology with long pay-back periods, and therefore the risks of policy uncertainty can constitute a significant barrier to its deployment. As such, effective policies and regulatory frameworks should be formulated to reduce or even eliminate the risks associated with the installation of CSP projects (Kiefer & del Río, 2020; Omri et al., 2019). The Rwanda Energy Policy (REP) was prepared in 2015, in support of the country's long-term economic development agenda to ensure reliable, affordable and sustainable energy access for all Rwandan residents and industries (MININFRA, 2015). However, the policy does not include any explicit regulation or legislation related to solar energy in general, or CSP in particular, which obviously discourages private sector participation in large energy projects.

If CSP is to succeed in Rwanda, comprehensive and proper policies should be formulated detailing framework conditions such as communicating ambitious targets for solar energy development; and various specific instruments such as deployment support (including feed-in tariffs) and R&D support. Some regulations could be provided as well, such as permit and planning processes and access to the grid.

### Conclusion

This study presented the techno-economic potential of CSP technologies in Rwanda, including the solar tower (STPP) and parabolic trough (PTPP). A 100 MW power plant was modelled for each technology in Nyanza and Kayonza,

where solar resources are relatively high. The significance of this study resides in the fact that no research has been done on CSP technologies in Rwanda and the study is in line with the government's efforts to increase the share of renewables in the power generation mix. Suitable designs for both technologies were modelled and it was found that an SM of 1.8 with a TES size of 8 h and an SM of 3.2 with a TES size of 6 h, were the optimal configurations for the STPP and PTPP, respectively. The corresponding LCOE values could then be minimized at 11.33 ¢/kWh and 12.92 ¢/kWh for the optimized STPP design at Nyanza and Kayonza, respectively, while they were 10.51 ¢/kWh and 11.30 ¢/kWh for the optimized PTPP design.

Then, a sensitivity analysis was carried out to examine the uncertainty in the technical and economic performance when parameters such as solar multiple (SM), storage size, debt interest rate and discount rate are changed. The results indicated that both the TES and solar field size affect significantly the technical and economic performance of both technologies. Comparing both technologies at the same plant configurations, it was found that the STPP perform better at smaller SM, while they are more suitable at higher TES sizes than the PTPP. The analysis also showed that the economic feasibility of CSP in Rwanda is hugely dependent on the financial parameters. The LCOE of the CSP project is largely increased with the increase of the debt interest rate, while the project is economically viable only when the discount rate varies between 10 and 24%.

Moreover, this study highlighted several barriers hampering the deployment of CSP in Rwanda and proposed some policy recommendations. Not only proper financing conditions are necessary but also effective policies and regulations are needed to ensure viable and competitive CSP projects in the country. The findings of this study may provide crucial science-based information to policymakers in further contemplating the deployment of CSP technologies. Future similar studies are suggested to evaluate other required factors for CSP feasibility, such as land requirement, access to water resources and grid infrastructure.

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## Declarations

**Conflict of interest** The authors declare that they have no conflict of interest.

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